



Indian Institute of Technology Gandhinagar

MA 104: ODE

Semester-II, Academic Year: 2025-26

Tutorial - 3

1. Verify that $y = x^2 \sin x$ and $y = 0$ are both solutions of the initial value problem (IVP)

$$x^2 y'' - 4xy' + (x^2 + 6)y = 0, \quad y(0) = 0, \quad y'(0) = 0.$$

Does this contradict the uniqueness of solutions for an IVP?

2. Let (x_0, y_0) be an arbitrary point in the plane and consider IVP

$$y' = y^2, \quad y(x_0) = y_0.$$

Show that $f(x, y) = y^2$ and $\partial f / \partial y$ are continuous on the entire plane. Find the intervals for the (unique) solutions through the points $(0, 0)$ and $(0, 1)$, respectively.

3. What does Picard's theorem say about the existence and uniqueness of solutions of

$$y' = \frac{3}{2}y^{1/3}, \quad y(0) = 0?$$

Show that this IVP has uncountably many solutions.

4. Consider the IVP

$$y' = \sqrt{y} + 1, \quad y(0) = 0, \quad x \in [0, 1].$$

Show that $f(x, y) = \sqrt{y} + 1$ does not satisfy a Lipschitz condition in any rectangle containing the origin, but that the solution is nevertheless unique.

Remark. If an IVP with f continuous (not necessarily Lipschitz) has more than one solution, then it has uncountably many solutions. This is known as *Kneser's Theorem*.

5. Use Picard's method of successive approximations to solve the following initial value problems and compare these results with the exact solutions:

(i) $y' = 2\sqrt{x}, \quad y(0) = 1$

(ii) $y' + xy = x, \quad y(0) = 0$

6. Consider the IVP

$$xy' - y = 0, \quad y(x_0) = y_0.$$

Solve it for different values of x_0 and y_0 . Does the result contradict Picard's theorem?

7. Find an interval on which the solution exists.

(a) $y' = 2 \sin(3xy), \quad y(0) = y_0.$

(b) $y' = \frac{e^{y^2} - 1}{1 - x^2 y^2}, \quad y(-2) = 1.$

8. Reduce to first order and solve, showing each step in detail.

(a) $y'' + (y')^3 \sin y = 0$

- (b) $y'' + \left(1 + \frac{1}{y}\right) (y')^2 = 0$
 (c) $xy'' + 2y' + xy = 0, \quad y_1 = \frac{\cos x}{x}$
 (d) $xy'' - 5xy' + 9y = 0, \quad y_1 = x^3$

9. Find an ODE for the given two independent solutions:

- (a) $e^{2.6x}$ and $e^{-4.3x}$
 (b) $\cos(2\pi x)$ and $\sin(2\pi x)$
 (c) $e^{-\sqrt{5}x}$ and $xe^{-\sqrt{5}x}$
 (d) $e^{(-2+i)x}$ and $e^{(-2-i)x}$

10. Solve the IVP. Show that your answer satisfies the ODE as well as the initial conditions.

- (a) $y'' + y' - 6y = 0, \quad y(0) = 10, \quad y'(0) = 0$
 (b) $4y'' - 4y' - 3y = 0, \quad y(-2) = e, \quad y'(-2) = -\frac{e}{2}$
 (c) $y'' - k^2y = 0 \ (k \neq 0), \quad y(0) = 1, \quad y'(0) = 1$
 (d) $8y'' - 2y' - y = 0, \quad y(0) = -0.2, \quad y'(0) = -0.325$

11. Solve $y'' - y = 0$ for the initial conditions $y(0) = 1, \quad y'(0) = -1$. Then change the initial conditions to $y(0) = 1.001, \quad y'(0) = -0.999$ and explain why this small change of 0.001 at $t = 0$ causes a large change later, e.g. 22 at $t = 10$. This is instability: a small initial difference in setting a quantity (a current, for instance) becomes larger and larger with time t . This is undesirable.

12. Find the harmonic motion $y(t) = A \cos \omega_0 t + B \sin \omega_0 t$ that starts from y_0 with initial velocity v_0 . Graph or sketch the solutions for $\omega_0 = \pi, y_0 = 1$ and various v_0 of your choice on common axes. At what t -values do all these curves intersect? Why?

13. Solve and graph the solution.

- (a) $x^2y'' + xy' + 9y = 0, \quad y(1) = 0, \quad y'(1) = 2.5$
 (b) $x^2y'' + 3xy' + y = 0, \quad y(1) = 3.6, \quad y'(1) = 0.4$
 (c) $x^2y'' + xy' + y = 0, \quad y(1) = 1, \quad y'(1) = 1$
 (d) $9x^2y'' + 3xy + y = 0, \quad y(1) = 1, \quad y'(1) = 0$

14. We present another method to obtain the second linearly independent solution of $y'' + ay' + by = 0$ when the auxiliary equation has real and equal roots.

- (a) For $m_1 \neq m_2$, verify that

$$y = \frac{e^{m_1 x} - e^{m_2 x}}{m_1 - m_2}$$

is a solution of $y'' - (m_1 + m_2)y' + m_1 m_2 y = 0$.

- (b) Keeping m_2 fixed, use L'Hospital's rule to find the limit of the solution in part (a) as $m_1 \rightarrow m_2$.
 (c) Verify that the limit obtained in part (b) satisfies the differential equation obtained from the equation in part (a) by replacing m_1 by m_2 .